



**SWAMINARAYAN
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THE RESERVOIR OF ALL KNOWLEDGE

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❖ **OSI MODEL**

The OSI (Open Systems Interconnection) Model describes how networked systems communicate. It is a theoretical framework for mapping how applications and devices link and send data over a web of connections.

The OSI Model is called a “reference model” because it is rarely used in practical applications. Instead, it is most helpful in describing network protocols and services. The OSI Model is most often used to understand the theories behind networked communication concepts and elements. Some use it to troubleshoot problems, narrowing a potential issue to a specific layer of the model.

OSI layers are, indeed, fundamental to the OSI Model. The framework is a stack of seven layers, each performing a specific function, yet working together to transmit data from one networked device to another. The 7 layers of OSI model ensure the interoperability of different devices and technologies from various vendors.

- [The Physical Layer](#)
- [The Data Link Layer](#)
- [The Network Layer](#)
- [The Transport Layer](#)
- [The Session Layer](#)
- [The Presentation Layer](#)
- [The Application Layer](#)

OSI Model

DATA	LAYER	FUNCTIONS
DATA	APPLICATION	Network process to application
DATA	PRESENTATION	Data representation and encryption
DATA	SESSION	Interhost communication
SEGMENTS	TRANSPORT	End -to-end connection and reliability
PACKETS	NETWORK	Path Determination and IP (local addressing)
FRAMES	DATA	MAC and LLC (physical addressing)
BITS	PHYSICAL	Media signal and transmission



7 Layers of the OSI Model



7. Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS



6. Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG



5. Session

- Synch & send to port
- API's, Sockets, WinSock



4. Transport

- End-to-end connections
- TCP, UDP



3. Network

- Packets
- IP, ICMP, IPSec, IGMP



2. Data Link

- Frames
- Ethernet, PPP, Switch, Bridge



1. Physical

- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters

1. Physical Layer

The first OSI model layer describes the physical connections between devices in a network. Electrical, optic, or electromagnetic signal data moves from device to device through the physical infrastructure defined by this layer.

Elements of the OSI layer model's physical layer include:

- Electrical, mechanical, and physical systems, parts, and devices
 - Specifications such as cable size, signal frequency, voltages, etc.
 - How a network is configured, such as a bus, with devices arranged in a line with many stops, a star, with a central device surrounded by an array of devices, a ring with devices in a connected circle, a mesh of woven interconnections, and more
 - Communication modes such as one-way simplex, two-way full duplex, or half duplex, with messages going back and forth, but only one direction at a time
 - Data transmission performance, such as bit rate, referring to how much data is moving through a connection at a given time, and bit synchronization to align the clocks of each device for accurate data transmission.
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- Modulation, switching, and interfacing with the physical transmission medium
Common protocols for connections, including Wi-Fi, Ethernet, and others
Hardware, including networking devices, antennas, cables, modems, and intermediate devices such as repeaters and hubs

2. The Data Link Layer

The second OSI model layer describes data transmission between network devices. Data is sent in packets within a frame of special bit patterns or codes that indicate the beginning and end of each packet. These packets move through switches that route information through the physical layer from one physical location to another.

The OSI data link layer has two sublayers:

- The Logical Link Control (LLC) sublayer manages flow and error controls to ensure accurate data transmission between the network devices.
- The Media Access Control (MAC) sublayer manages access and permissions for transmitting data between devices. The function of this sublayer is to manage which device controls a channel, moment to moment.

3. The Network Layer

The third of the OSI layers organizes and transmits data between multiple networks. Network layer hardware includes routes, bridge routers, 3-layer switches, and protocols such as Internet (IPv4) Protocol version 4 and Internet Protocol version 6 (IPv6).

This layer routes data via the shortest or fastest physical path, working around traffic controls, congestion, broken links, service priority, and more. It handles addressing logic to distinguish between the source and destination networks. It divides data into packets to send and then reassembles them at their destination.

4. The Transport Layer

This layer deals with sending and delivering of complete and reliable data from one device to another through a network or between networks. Common transport layer protocols include the Transmission Control Protocol (TCP) for connection-oriented data transmission and the User Datagram Protocol (UDP) for connectionless data transmission.

Some of the essential functions in this layer include:

- Error control, flow control, and congestion control are ways to keep track of data packets, check for errors and duplication, and then to resend if there is an error or failure.
- Service-point addressing ensures that data is delivered to the correct protocol, identified by a port number.
- Packet segmentation and reassembly are processes for dividing data and sending it sequentially, then rechecking it at its destination for integrity and accuracy.

5. The Session Layer

This OSI Model layer concerns itself with managing session links between network devices with a specific beginning and end. The essential functions include establishing a link to start the session, authenticating senders and receivers, authorizing communications between devices and apps, maintaining the session, and terminating the connection, which are the key functions.

A common session type is when an internet user visits and browses a website for a specific period.

An essential concept in this layer is synchronization, in which checkpoints during the session ensure a coordinated data flow that is free of unplanned breaks or data loss.

Common OSI session layer protocols include:

- Remote procedure call protocol (RPC)
- Point-to-Point Tunneling Protocol (PPTP)
- Session Control Protocol (SCP)
- Session Description Protocol (SDP)

6. The Presentation Layer

Data flows as packets of code, often encrypted, through networks. Presenting it in a useful format is the focus of the sixth OSI model layer. The presentation layer deals with:

- Data conversion
- Character code translation
- Data compression
- Encryption and decryption

To send text across a network, the characters of the alphabet convert via a character coding system, such as the American Standard Code for Information Interchange (ASCII) or Extended Binary Coded Decimal Interchange Code (EBCDIC) that is then encrypted and compressed and sent over the network. On the receiving end, the process reverses. Different kinds of data get translated into different format codes.

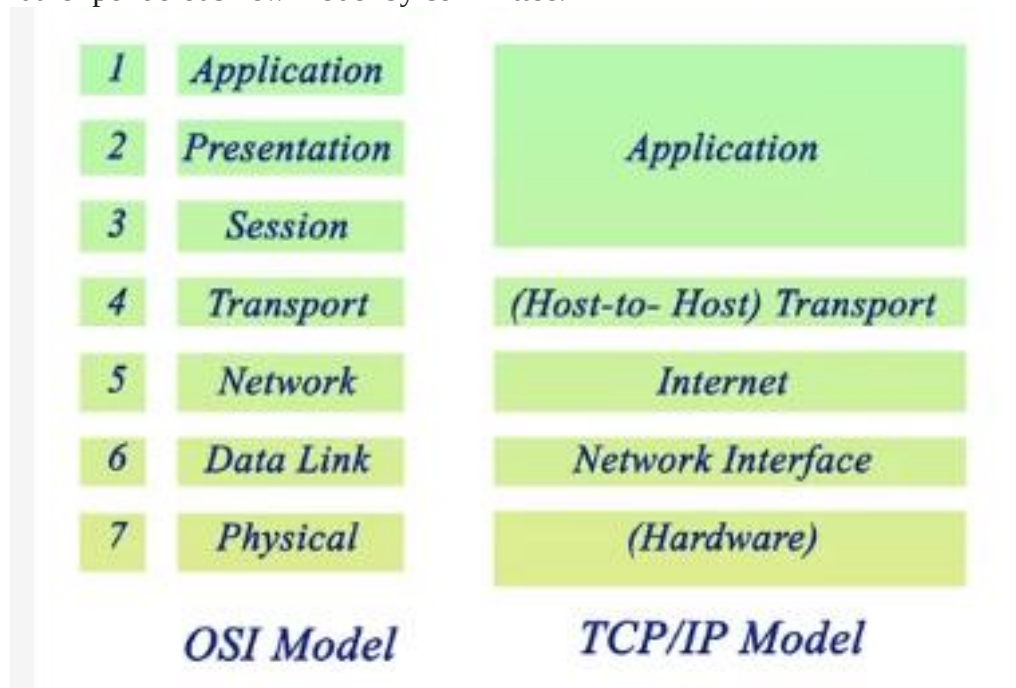
7. The Application Layer

The top layer of the OSI 7 layer model is the application layer. It is how a user application, such as a website, browser, email, instant messaging, file transfers, or voice-over IP, interfaces with the network. Think of it as the window for accessing the network to send or display data, which can be anything from a picture of your pet cat to a database of statistics to a voice message.

It facilitates networking requests, determines resource availability, synchronizes communication, and manages application-specific networking requirements. The application layer also identifies constraints at the application level, such as those associated with authentication, privacy, quality of service, networking devices, and data syntax.

OSI vs. TCP/IP

The OSI vs. TCP/IP debate never really gained traction in the industry. Vendors had already invested significant resources in TCP/IP products in the ten-plus years before the launch of the OSI. They had no incentive to manage the interoperability issues with the vast choices of protocols and specifications offered by the OSI model. Additionally, the creation of the OSI Model was driven by [European political and economic interests](#) that didn't want to be dependent on technologies and equipment developed by American companies. Hence, a non-governmental organization, the International Organization for Standardization, designed the rather ponderous new model by committee.



❖ Differences between OSI and TCP/IP model

	OSI	TCP/IP
1. Layer Structure	OSI has 7 distinct layers, each with a clearly defined function.	TCP/IP has 4 layers with some combined responsibilities.
2. Protocol Dependency	OSI is protocol-independent , meaning it doesn't specify protocols — it's more a guideline.	TCP/IP is protocol-specific , with defined standards like TCP, IP, FTP, etc.
3. Modularity and Flexibility	OSI's strict modular design makes it easier to troubleshoot specific layers.	TCP/IP's loose layering allows faster and more flexible integration of protocols.
4. Real-World Adoption	OSI is mostly used for teaching and conceptual clarity.	TCP/IP is the backbone of the modern internet — globally adopted.
5. Development Sequence	OSI model was developed before the protocols .	TCP/IP protocols were developed before the model , which led to a more practical approach.
6. Reliability	OSI assigns reliability to the Transport layer only.	TCP/IP allows reliability to be handled at multiple layers , mostly via TCP in the Transport layer.

❖ TCP/IP MODEL

The **TCP/IP Model** (Transmission Control Protocol/Internet Protocol) is a *practical networking framework* developed in the 1970s by DARPA for reliable communication between computers. Unlike the OSI model, it's based directly on working protocols used in the real world. TCP/IP consists of **four layers**, each handling specific functions:

1. Application Layer

- **Purpose:** Interfaces with user applications.
- **Responsibilities:**
 - Data generation (e.g., web browsing, email, file transfer)
 - Protocols:
 - **HTTP/HTTPS** (web)
 - **FTP** (file transfer)
 - **SMTP/POP/IMAP** (email)
 - **DNS** (domain name resolution)
- **Corresponding OSI Layers:** Combines OSI's Application, Presentation, and Session layers.

2. Transport Layer

- **Purpose:** Provides reliable or unreliable data delivery.
- **Key Protocols:**
 - **TCP** (Transmission Control Protocol): Reliable, connection-oriented
 - **UDP** (User Datagram Protocol): Faster, connectionless
- **Functions:**
 - Error detection and correction
 - Flow control
 - Data segmentation and reassembly

3. Internet Layer

- **Purpose:** Handles routing and addressing.
- **Key Protocols:**
 - **IP** (Internet Protocol): Addressing and delivery of packets
 - **ICMP** (for diagnostics — ping)
 - **ARP** (Address Resolution Protocol)
- **Functions:**
 - Logical addressing
 - Routing across networks
 - Packet forwarding

4. Link Layer (or Network Access Layer)

- **Purpose:** Manages physical transmission of data.
- **Components:**
 - Device drivers
 - Network interface cards (NICs)
 - MAC addressing
- **Functions:**
 - Frame formatting
 - Error detection at hardware level
 - Access control (who can send data on the medium)